

A red parallelogram tilted to the right, containing the text "Projects ada" in white.

Projects
ada

EPFL

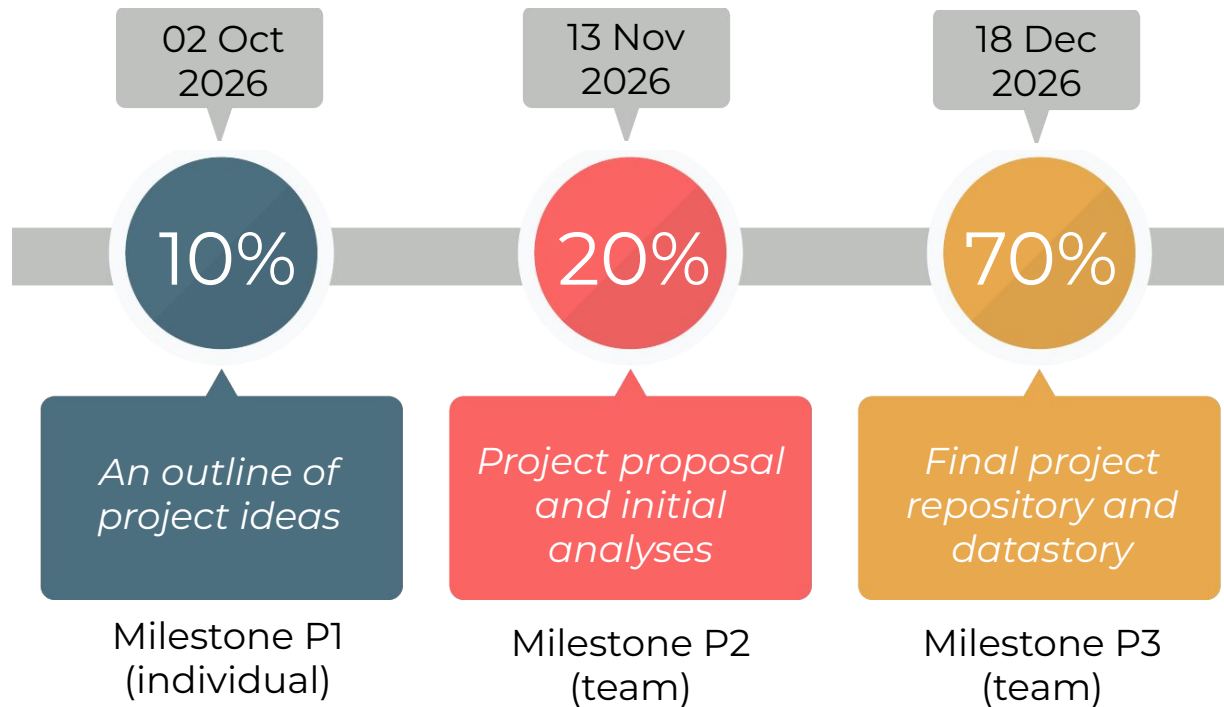
Project

Weight: 30%

All info available here!

Overview

Three milestones: first glimpse, proposal, execution



All the dates have deadline at 23:59

Datasets

Provided datasets are available here:

https://go.epfl.ch/ada26_datasets

Wikispeedia

Human navigation behavior data on a subset of Wikipedia, including:

- Human navigation paths
- Wikipedia hyperlink graph
- Plain-text content of Wikipedia articles

Modalities: Text / Graph / Navigation behavior

Generic and versatile

YouNiverse*

Metadata about Youtube channels and videos

Collected for 14 years from 2005 – 2019, including:

- Channel-level time series: weekly subscriber and view counts
- Co-commenting graph

Modalities: Timeseries / Graph

[Helpful dataset overview](#)

Stock Market Dataset

Historical daily prices for all tickers currently trading on NASDAQ (up to March 2020)

- Temporal data for each of the NASDAQ traded tickers (5884 stocks and 2165 ETFs)
- For each, values on: date, open, high, low, close, and volume

Modalities: Numeric data / Categorical / Time series

[Helpful Dataset Overview](#)

Reddit Hyperlink Network

Subreddit-to-subreddit hyperlink network extracted from posts that create hyperlinks from one subreddit to another

- Network is directed, signed, temporal, and attributed
- Each item includes:
 - Source and target subreddit
 - ID + timestamp of source subreddit
 - Post label: 1 if tone towards target is positive/neutral, -1 if it is negative
 - Post properties: vector representing the text properties of the post
- Can be combined with embedding vectors dataset

Modalities: Graphs / Categorical / Temporal / Text

[Helpful Dataset Overview](#)

New Yorker Caption Contest

384 cartoon caption contests from New Yorker with ~1.4M captions and ~183M ranked votes

Aligned metadata includes:

- Image locations, descriptions, wikipedia entities, caption ratings

Modalities: Text / Numeric data / Image

Can be combined with other sources (e.g. Wikipedia)

[Helpful Dataset Overview](#)

Multi-tissue COVID-19 Atlas

Collection of single or multi-omics datasets from 14 studies

- Tissue is primarily PBMC (other tissues: kidney, nasal epithelium)
- Focus: PBMC study
 - Contains 640k cells, 24k genes, 192 surface protein markers
 - Interesting metadata: Collection Day, Disease outcome, Age, Sex, Cell type, Smoker, Worst patient status

Modalities: Tabular numeric data (raw gene & protein counts)

[Helpful Dataset Overview](#)

Visual Genome

Connects structured image concepts to language

Over 108K images, each annotated with:

- Objects, attributes and pairwise relationships between objects
- Region descriptions and question-answer pairs
- On average: 35 objects, 26 attributes, 21 relationships per image

Modalities: Text / Image / Graph

[Helpful dataset overview](#)

Cyclistic Bike Share Dataset

Bike-share trip data from June 27, 2013 to December 31, 2024

Each trip is anonymized and includes:

- Trip start and end day and time
- Trip start and end station
- Rider type: Member (or Subscriber), Single Ride, Day Pass

Modalities: Text / Numeric data / Coordinates

[Helpful dataset overview](#)

Milestone 1

To be done individually, due 2nd Oct 23:59

No questions will be answered on Ed 24h before the deadlines!

1. Read the paper(s) relevant (Column G) to the dataset (fine if you don't fully grasp the technical details about the method)
2. Play around with the dataset
3. Propose **three** bold and creative ideas (non-binding)

Deliverable: Outline of project ideas (up to 500 words)

Interesting examples from past years

Milestone 2

To be done as a team, due 13th Nov 23:59

1. A detailed project proposal (up to 1000 words)
2. A well-organised repository with initial analyses and data handling pipelines

Milestone 3

To be done as a team, due 18th Dec 23:59

1. The final repository: complete with all the analysis and findings
2. Provide a data story
 - 2.1. Not a loose collection of results
 - 2.2. Examples from [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), [2022](#), [2023](#), [2024](#)

Usage of LLMs

For Milestone 1, LLMs are not permitted.

For Milestones 2 and 3, please explicitly state in your report where LLMs were used.

Thanks for your attention!